

$$m = 0.5 \text{ kg}$$

13 A A is correct. Since $\Delta U = Q + W \Rightarrow \Delta U$ can be increased by Q and/or W.

B is wrong because internal energy U = sum of ke and pe of molecules of a system. Only ke depends on temperature but pe depends on intermolecular forces of attraction and separation between molecules.

C and D are wrong statements, same explanation as B.

Note: if the system is an ideal gas, then B, C and D are correct statements.

14 B Amplitude = 5.0 cm